REDOX REACTIONS

- **Redox Reactions** (oxidation-reduction reactions) involve transference of an electron(s) between reactants.

MEMORY TOOL

LEO the lion goes **GER**

L ose E lectrons O xidation	G ain E lectrons R eduction
☐ Become more positive in charge.	☐ Become more negative in charge.
☐ Oxidation number increases.	☐ Oxidation number decreases.



Reduction and **Oxidation**: the reactions of oxidizing and reducing agents.

- ☐ **Oxidizing Agent**: The element or compound that is reduced.
- ☐ **Reducing Agent:** The element or compound that is oxidized.

EXAMPLE: Consider the following reaction below when solid lithium reacts with the zinc ion:

$$2\ Li\ (s) + Zn^{2+}\ (aq) \longrightarrow 2\ Li^{+}\ (aq) + Zn\ (s)$$

Which reactant is undergoing **oxidation** and which reactant is undergoing **reduction**?

Li is oxidized (0 → +1, loses e⁻) — Li is the reducing agent

Zn^{2+} is reduced (+2 → 0, gains e⁻) — Zn^{2+} is the oxidizing agent

PRACTICE: Which element is being reduced in the following reaction?

$$Cr_2O_7^{2-} + 3\ HNO_2 + 5\ H^{+} \longrightarrow 2\ Cr^{3+} + 3\ NO_3^{-} + 4\ H_2O$$

Cr (in $Cr_2O_7^{2-}$) is reduced: +6 → +3

(N in HNO_2 is oxidized: +3 → +5)

PRACTICE: Identify the oxidizing agent and reducing agent from the following redox reaction.

$$Ba\ (s) + Cl_2\ (g) \longrightarrow BaCl_2\ (aq)$$

Oxidizing agent: Cl_2 (0 → −1) Reducing agent: Ba (0 → +2)

PRACTICE: Which element is oxidized and which is reduced in the following reaction?

$$Hg\ (aq) + HgCl_2\ (aq) \longrightarrow Hg_2Cl_2\ (aq)$$

Hg (0 → +1) is oxidized; Hg in $HgCl_2$ (+2 → +1) is reduced

PRACTICE: Which of the following represents an oxidation-reduction reaction?

I. $PCl_3\ (aq) + Cl_2\ (g) \longrightarrow PCl_5\ (aq)$

II. $2\ AgNO_3\ (aq) + Cu\ (s) \longrightarrow Cu(NO_3)_2\ (aq) + 2\ Ag\ (s)$

III. $CO_2\ (g) + 2\ LiOH\ (aq) \longrightarrow Li_2CO_3\ (aq) + H_2O\ (l)$

IV. $FeCl_2\ (aq) + 2\ NaOH\ (aq) \longrightarrow Fe(OH)_2\ (aq) + 2\ NaCl\ (aq)$

a) I, II, III, and IV

b) I, II, and III

c) III

d) I and II

e) IV

Answer: d) I and II

In each of the following equations, state which reactant has been oxidised:

i $PbO + H_2 \rightarrow Pb + H_2O$

H_2

ii $CO + Ag_2O \rightarrow 2Ag + CO_2$

CO

iii $2Mg + CO_2 \rightarrow 2MgO + C$

Mg

In each of the following equations, state which reactant has been reduced:

i $5CO + I_2O_2 \rightarrow 5CO_2 + I_2$

I_2O_5 (iodine oxide; I +5 → 0)

ii $2H_2S + SO_2 \rightarrow 3S + 2H_2O$

SO_2 (S +4 → 0)

iii $CH_2{=}CH_2 + H_2 \rightarrow CH_3CH_3$

$CH_2{=}CH_2$ (ethene)

Oxidation number rules

IMPORTANT

The more positive (or less negative) the oxidation number, the more oxidised a substance is. The less positive (or more negative) the oxidation number, the more reduced a substance is. Take care with negative oxidation numbers: Remember that an ox. no. of −3 is less oxidised / more reduced than an ox. no. of −2.

We can deduce the oxidation number of any atom or ion by oxidation number rules. It is important to note that an oxidation number refers to a *single atom* in a compound.

1 The oxidation number of any uncombined element is zero. For example, the oxidation numbers of each atom in S_8, Cl_2 and Zn is zero.

2 In compounds many atoms or ions have fixed oxidation numbers:

- Group 1 elements are always +1
- Group 2 elements are always +2
- fluorine is always −1
- hydrogen is +1 (except in metal hydrides such as NaH, where it is −1)
- oxygen is −2 (except in peroxides, where it is −1, and in F_2O, where it is +2).

3 The oxidation number of an element in a monatomic ion is always the same as the charge. For example, Cl^- is −1, Al^{3+} is +3.

4 The sum of the oxidation numbers in a compound is zero.

5 The sum of the oxidation numbers in an ion is equal to the charge on the ion.

6 In either a compound or an ion, the more electronegative element is given the negative oxidation number.

This exercise will familiarise you with the use of **oxidation number** rules to deduce the oxidation state of atoms or ions in compounds.

> **TIP**
>
> Check that you know the fixed oxidation numbers of particular atoms or ions before starting this exercise.

a Copy and complete the following sentences.

i The sum of the oxidation numbers in a compound is zero.

ii The sum of the oxidation numbers in an ion is equal to the charge on the ion.

iii The oxidation number of fluorine in compounds is −1.

iv The oxidation number of oxygen in compounds is −2, except in peroxides where it is −1

v The total oxidation number of the sulfur and the four oxygen atoms in SO_4^{2-} is −2.

b Fill in the gaps in the procedure given below to find the oxidation number of Fe in Fe_2O_3.

The sum of the oxidation numbers of all the atoms in Fe_2O_3 is 0.

Each O atom has an oxidation number of −2.

The oxidation number for three O atoms is −6.

The oxidation number for two Fe atoms is +6.

So each Fe atom has an oxidation number of +3.

c Fill in the gaps in the procedure given below to find the oxidation number of N in the NO_3^- ion.

The sum of the oxidation numbers of all the atoms in the NO_3^- ion is −1.

Each O atom has an oxidation number of −2.

The oxidation number for three O atoms is −6.

The oxidation number for the N atom is +5.

d Deduce the oxidation numbers of the atoms that are underlined.

i $\underline{Cr}_2O_3$ +3

ii $Sr\underline{Br}_2$ −1

iii $\underline{S}O_3$ +6

iv $\underline{As}_2O_5$ +5

v $H\underline{Cl}O_4$ +7

vi $\underline{P}O_4^{3-}$ +5

vii $\underline{S}O_3^{2-}$ +4

OXIDATION NUMBERS I

a) Work out the oxidation number of *nitrogen* in each of the following substances.

(i) NO ..... +2 .....

(ii) NO_2 ..... +4 .....

(iii) NF_3 ..... +3 .....

(iv) N_2O_5 ..... +5 .....

(v) NH_3 ..... −3 .....

(vi) HNO_3 ..... +5 .....

(vii) N_2 ..... 0 .....

(viii) N_2O ..... +1 .....

[8]

b) Work out the oxidation number of *chlorine* in each of the following substances.

(i) Cl_2O ..... +1 .....

(ii) Cl_2O_7 ..... +7 .....

(iii) ClO_2 ..... +4 .....

(iv) KCl ..... −1 .....

[4]

c) Work out the oxidation number of the named elements in the following *ions:*

(i) Chlorine in ClO^- ..... +1 .....

(ii) Chromium in CrO_4^{2-} ..... +6 .....

(iii) Aluminium in AlF_6^{3-} ..... +3 .....

(iv) Oxygen in OH^- ..... −2 .....

(v) Phosphorus in PO_4^{3-} ..... +5 .....

(vi) Chlorine in ClO_4^- ..... +7 .....

(vii) Chromium in $Cr_2O_7^{2-}$ ..... +6 .....

(viii) Copper in CuF_4^{2-} ..... +2 .....

(ix) Oxygen in O_2^{2-}. ..... −1 .....

(x) Nitrogen in NO_2^- ..... +3 .....

[10]

d) Complete the following names by adding the appropriate Roman numeral:

(i) SO_4^{2-} sulfate(VI) ion

(ii) SO_3^{2-} sufate(IV) ion

(iii) CuO copper(II) oxide

(iv) Cu_2O copper(I) oxide

(v) $NaClO_3$ sodium chlorate(V)

[5]

Faisal has written the following notes on redox reactions in preparation for his AS exams. However there are a few mistakes, many of which are commonly seen in exam answers. Help Faisal learn from his mistakes by correcting the errors so that he has an accurate set of notes to revise from.

Redox reactions (10 marks)

Oxidation is the loss of electrons; the oxidation state decreases ✗ should be: INCREASES

Reduction is the gain of electrons; the oxidation state increases ✗ should be: DECREASES

An **oxidising agent** oxidises something and is itself reduced

A **reducing agent** reduces something and is itself oxidised

reduction

✗ +2 ✗ 0 ✗ +4

+3 +1 +1 +2

e.g. $Fe_2O_3 + 3\ CO \rightarrow 2\ Fe + 3\ CO_2$

oxidation

Fe_2O_3 is **reduced** and is therefore acting as a **reducing agent**;

CO is **oxidised** and is therefore acting as an **oxidising agent**.

✗ OXIDISING

✗ REDUCING

Rules for balancing redox reactions

1. Use oxidation states to identify the species undergoing oxidation or reduction.

   $Fe_2O_3 \rightarrow Fe$ reduction $CO \rightarrow CO_2$ oxidation

2. Balance the atoms undergoing oxidation or reduction.

   $Fe_2O_3 \rightarrow 2\ Fe$ $CO \rightarrow CO_2$

3. Balance the oxygen atoms by adding H_2O.

   $Fe_2O_3 \rightarrow 2\ Fe + 3\ H_2O$ $H_2O + CO \rightarrow CO_2$

4. Balance the hydrogen atoms by adding H^+.

   $6\ H^+ + Fe_2O_3 \rightarrow 2\ Fe + 3\ H_2O$ $H_2O + CO \rightarrow CO_2 + 2\ H^+$

5. Balance the electrons.

   $6\ e^- + 6\ H^+ + Fe_2O_3 \rightarrow 2\ Fe + 3\ H_2O$ $2\ e^- + H_2O + CO \rightarrow CO_2 + 2\ H^+$ ✗ (1)

6. Multiply either half equation to make sure that the no. of electrons in each half equation match.

   $6\ e^- + 6\ H^+ + Fe_2O_3 \rightarrow 2\ Fe + 3\ H_2O$ $6\ e^- + 3\ H_2O + 3\ CO \rightarrow CO_2 + 6\ H^+$ ✗ (2)

7. Combine to get the full redox equation.

   $6\ e^- + 6\ H^+ + Fe_2O_3 + 3\ H_2O + 3\ CO \rightarrow 2\ Fe + 3\ H_2O + 3\ CO_2 + 6\ H^+ + 6\ e^-$

8. Cancel out anything that appears on both sides of the equation.

   ~~$6\ e^-$~~ + ~~$6\ H^+$~~ + Fe_2O_3 + ~~$3\ H_2O$~~ + $3\ CO \rightarrow 2\ Fe$ + ~~$3\ H_2O$~~ + $3\ CO_2$ + ~~$6\ H^+$~~ + ~~$6\ e^-$~~

   $Fe_2O_3 + 3\ CO \rightarrow 2\ Fe + 3\ CO_2$

Corrections: C in CO = +2, Fe = 0, C in CO_2 = +4; Fe_2O_3 is the oxidising agent, CO is the reducing agent.

(1) Step 5: $H_2O + CO \rightarrow CO_2 + 2H^+ + 2e^-$ (electrons go on the product side for oxidation)

(2) Step 6: $3H_2O + 3CO \rightarrow 3CO_2 + 6H^+ + 6e^-$ (multiply every species by 3, incl. CO_2)

Balance the half equations by balancing the atoms and adding

	Electrons	H^+	H_2O
1. Mg		$\rightarrow$ Mg^{2+}	$+ 2e^-$
2. Cl_2	$+ 2e^-$	$\rightarrow$ $2\ Cl^-$	
3. H_2O_2		$\rightarrow$ O_2	$+ 2H^+ + 2e^-$
4. SO_4^{2-}	$+ 4H^+ + 2e^-$	$\rightarrow$ SO_2	$+ 2H_2O$
5. CH_3CH_2OH	$+ H_2O$	$\rightarrow$ CH_3COOH	$+ 4H^+ + 4e^-$
6. CH_3CH_2OH		$\rightarrow$ CH_3CHO	$+ 2H^+ + 2e^-$
7. $Cr_2O_7^{2-}$	$+ 14H^+ + 6e^-$	$\rightarrow$ $2\ Cr^{3+}$	$+ 7H_2O$
8. MnO_4^-	$+ 8H^+ + 5e^-$	$\rightarrow$ Mn^{2+}	$+ 4H_2O$
9. H_2S		$\rightarrow$ S	$+ 2H^+ + 2e^-$
10. $^2IO_3^-$	$+ 12H^+ + 10e^-$	$\rightarrow$ I_2	$+ 6H_2O$

BALANCING REDOX REACTIONS: ACIDIC SOLUTIONS

- **Balancing Redox Reactions** requires a new approach that accounts for the transfer of electrons between reactants.
  - For **Acidic Redox Reactions** we generally have the presence of the H^+ ion.
  - Redox reactions not only balance the atoms of elements, but also charge and electrons.

Half-Reactions

- Balancing a redox reaction begins with identifying its *half reactions*.
  - **Half Reaction:** Either the oxidation or reduction reaction portion of a redox reaction.

Balancing Acidic Redox Reactions

EXAMPLE: Balance the following redox reaction if it is found to be in an acidic solution.

$$NO_2^-(aq) + Cr_2O_7^{2-}(aq) \longrightarrow Cr^{3+}(aq) + NO_3^-(aq)$$

STEP 1: Break the full redox reaction into 2 half reactions.

- Focus on the elements that are not **oxygen** or **hydrogen** to determine the 2 half-reactions.

Ox: $NO_2^- \rightarrow NO_3^-$ Red: $Cr_2O_7^{2-} \rightarrow Cr^{3+}$

STEP 2: For each half reaction, balance elements that are not **oxygen** or **hydrogen**.

$NO_2^- \rightarrow NO_3^-$ $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$

STEP 3: For each half reaction, balance the number **oxygens** by adding H_2O.

$NO_2^- + H_2O \rightarrow NO_3^-$ $Cr_2O_7^{2-} \rightarrow 2Cr^{3+} + 7H_2O$

STEP 4: For each half reaction, balance the number **hydrogens** by adding H^+.

$NO_2^- + H_2O \rightarrow NO_3^- + 2H^+$ $Cr_2O_7^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O$

STEP 5: Balance the overall charge by adding **electrons** to the more positive charged side of each half reaction.

- If the number of electrons of both half reactions differ then multiply to get the lowest common multiple.

$3 \times [NO_2^- + H_2O \rightarrow NO_3^- + 2H^+ + 2e^-]$ $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

STEP 6: Combine the half reactions and cross out *reaction intermediates*.

- **Reaction Intermediates**: Compounds that look the same, with one as a reactant and the other a product.

$Cr_2O_7^{2-}(aq) + 3NO_2^-(aq) + 8H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3NO_3^-(aq) + 4H_2O(l)$

1 Produce balanced equations for the following reactions:

a the reduction of iodate(V) ions, IO_3^-, to iodine, I_2, by iodide ions in acid solution

$IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

b the oxidation of NO_2^- to NO_3^- with manganate(VII) ions, MnO_4^-, in acid solution

$2MnO_4^- + 5NO_2^- + 6H^+ \rightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

c the reduction of VO_2^+ to V^{2+} with metallic zinc in acid solution

$2VO_2^+ + 3Zn + 8H^+ \rightarrow 2V^{2+} + 3Zn^{2+} + 4H_2O$

d the oxidation of $Fe(OH)_2$ to $Fe(OH)_3$ by oxygen.

$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3$

2 (Harder) Write balanced equations for the following reactions:

a the reduction of $Cr_2O_7^{2-}$ to Cr^{3+} with metallic magnesium in acid solution

$Cr_2O_7^{2-} + 3Mg + 14H^+ \rightarrow 2Cr^{3+} + 3Mg^{2+} + 7H_2O$

b the reduction of nitrate(V) to ammonia with metallic aluminium in alkaline solution

$3NO_3^- + 8Al + 18H_2O \rightarrow 3NH_3 + 8Al^{3+} + 27OH^-$

(or with aluminate: $3NO_3^- + 8Al + 5OH^- + 18H_2O \rightarrow 3NH_3 + 8[Al(OH)_4]^-$)

c the oxidation of $Cr(OH)_3$ to CrO_4^{2-} by H_2O_2 in alkaline solution.

$2Cr(OH)_3 + 3H_2O_2 + 4OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$

Tip: Balance with acid rule first and then add OH- to remove the H+ used. This forms water.

You must add OH - to BOTH sides of equation when you do this.

Q1 Iodide ions are stronger reducing agents than bromide ions. When they react with concentrated sulphuric acid a mixture of reduction products are formed from the sulphuric acid.

a) Combine two half- equations from those overleaf to deduce the full ionic equation for reduction of sulphate ions to sulphur by iodide ions.

$SO_4^{2-} + 8H^+ + 6e^- \rightarrow S + 4H_2O$ $2I^- \rightarrow I_2 + 2e^-$ (×3)

Overall: $SO_4^{2-} + 8H^+ + 6I^- \rightarrow S + 3I_2 + 4H_2O$

b) Combine two half- equations from those overleaf to deduce the full ionic equation for reduction of sulphate ions to hydrogen sulphide, H_2S, by iodide ions.

$SO_4^{2-} + 10H^+ + 8e^- \rightarrow H_2S + 4H_2O$ $2I^- \rightarrow I_2 + 2e^-$ (×4)

Overall: $SO_4^{2-} + 10H^+ + 8I^- \rightarrow H_2S + 4I_2 + 4H_2O$

Q2 Concentrated sulphuric acid may oxidise sulphur. Both of the reactants are converted into sulphur dioxide during the reaction. Deduce the full ionic equation for the reaction by writing the two half-equations involved and combining them together.

Ox: $S + 2H_2O \rightarrow SO_2 + 4H^+ + 4e^-$ Red: $SO_4^{2-} + 4H^+ + 2e^- \rightarrow SO_2 + 2H_2O$ (×2)

Overall: $S + 2SO_4^{2-} + 4H^+ \rightarrow 3SO_2 + 2H_2O$

Q3 Sulphite ions, SO_3^{2-}, often act as reducing agents in reactions; they are oxidised to sulphate ions, SO_4^{2-}.

a) Deduce the half-equation to show the oxidation of sulphite ions to sulphate ions.

$SO_3^{2-} + H_2O \rightarrow SO_4^{2-} + 2H^+ + 2e^-$

b) Iron(III) ions are reduced to iron(II) ions by sulphite ions. Deduce the half-equation for the reduction of iron(III) ions to iron(II) ions and combine the two half-equations to derive the overall ionic equation for the reduction of iron(III) ions by sulphite ions.

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ (×2)

Overall: $2Fe^{3+} + SO_3^{2-} + H_2O \rightarrow 2Fe^{2+} + SO_4^{2-} + 2H^+$